

An-Chi Liu

SOFTWARE ENGINEER · SYSTEM SOFTWARE & ROBOTIC & AUTONOMOUS DRIVING PROFESSION

Taiwanese, based in Tokyo, Japan (Permanent Resident)

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Work Experience

T2, Inc. (L4 Truck Autonomous Driving)

Senior Software Engineer

- Built a **simulation-based QA cluster platform** from scratch for running T2's **autonomous driving** simulation, evaluation, and **automated regression testing** tools, now adopted company-wide and running **50+ simulations daily**. Designed the system architecture and implemented the full stack with **Kubernetes** and **Docker** for **distributed simulation execution**, a **Golang** backend, and a **React** frontend, all built with **AI-driven development** practices. Also designed and built the underlying cluster hardware and networking.
- Designed and implemented an **OpenDrive-compatible mapping system** and an **OpenScenario-compatible scenario library**, enabling **closed-loop simulation** and **large-scale benchmarking** of the **autonomous driving planning stack** across diverse maps, scenarios, and **safety-critical edge cases**, with **evaluation metrics** and **dashboards** to quantify **planner behavior** and catch regressions.

Hybrid
Nov. 2025 – Present

Datra AI

Technical Co-founder

- Co-founded an AI startup for **Shopify** merchants; led end-to-end product development across infrastructure, system architecture, backend (**Golang**), frontend (**React**), and deployment on **AWS** with **Docker**, managing up to **2 developers**.
- Built an **AI system** integrating various **LLMs** (e.g., OpenAI, Claude), with optimized **prompt engineering** and retrieval-augmented generation (**RAG**) techniques.

Remote
Feb. 2025 – Sep. 2025

Mujin, Inc.

Backend Software Engineer

- Developed a universal **network device discovery service** in **C++** using **PcapPlusPlus** that auto-detects **Ethernet/IP** devices (robots, **industrial cameras**, etc.) on the network within seconds.
- Built a **real-time live streaming and playback system** for over **100** surveillance cameras with end-to-end latency under **50 ms**, using **FFmpeg**, **WebRTC**, and **Golang**.
- Maintained and extended production **real-time robot and warehouse control systems** (motion, sensors, and **manipulation** pipelines) written primarily in **C++** (with **Python**), using **ZMQ** for inter-process messaging and **Qt/QML** for operator UIs.
- Implemented new **web APIs** in **Golang** and **ZMQ APIs** in **C++** to support frontend projects and internal system services.

Tokyo, Japan
Sep. 2022 – May 2025

Education

National Yang Ming Chiao Tung University (NYCU)

M.S. in Computer Science and Engineering

- GPA: 4.11/4.3. Thesis: **Offworker: An Offloading Framework for Parallel Web Applications**, published in **WISE 2022** (top-tier conference).
- Received the **Best Master's Thesis Award** from the Institute of Information & Computing Machinery, Taiwan (2022; one of 13 awardees).

Hsinchu, Taiwan
Sep. 2019 – May. 2022

National Taiwan University (NTU)

B.S. in Bio-Industrial Mechatronics Engineering

Taipei, Taiwan
Sep. 2014 – Jun. 2019

Open-Source Projects

solvcon

- solvcon is a high-performance toolkit for solving partial differential equations, built with **C++** and **pybind11** bindings, with **61 stars** on GitHub.
- **The major contributor**, with contributions including a **2D graphic editor**, **AI agent for graphic**, **low-level NumPy-like array**, **polygon boolean operations and data structures**, and **drawing APIs** for the pilot GUI tool.
- Gave a talk at **PyCon JP 2024**: "Crafting Your Own NumPy: Do More in C++ and Make It Python".

PcapPlusPlus

- PcapPlusPlus is a multi-platform **C++** library for capturing, parsing, and crafting **network packets** across **Ethernet** and many other protocols, with **3k stars** on GitHub.
- **One of five core maintainers** of PcapPlusPlus since 2024.
- Major contributions include fixing performance issues, modernizing the **CI/CD** pipeline, adding new protocol support, and improving documentation.

Skills

Programming	Proficient: C++11/14/17 (real-time systems), Python 3, Experience: Golang, JavaScript, TypeScript, Node.js
Robotics/Simulation	Closed-loop simulation, OpenSCENARIO, OpenDRIVE, scenario/benchmark design, planning-stack evaluation, evaluation metrics/dashboards, regression testing, log replay, motion/manipulation pipelines, GPU clusters
Frameworks/Tools	Linux, CMake, GDB, Git, CI/CD, Docker, Kubernetes, AWS, PcapPlusPlus, ZMQ, FFmpeg, WebRTC, pybind11, GoogleTest, Qt/QML, REST API, GraphQL, React, etc.
AI Skills	Claude Code, Codex, LLM integration (OpenAI, Claude, etc.), Prompt engineering, RAG, MCP, etc.
Languages	Mandarin Chinese (Native), English (Business), Japanese (Limited Conversational)